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# WO-500-04 — Replace VALVE, BALL, FLOATING 1" P150 (1"-P150-HS-5011/A3)

| | | | |
|---|---|---|---|
| **Area** | A500 Pipe rack | **Task** | VR Valve replacement in kind |
| **Priority** | 3 (reliability) | **Issued by** | Facility owner (demonstration) |
| **Primary asset** | rack-bent:720 — Pipe rack bent at E 60'-0" — E 60'-0"  N 0.000"  EL 122'-0" | | |
| **Equipment** | — | **Lines** | 1"-P150-HS-5011 |
| **Isometrics** | ISO-1"-P150-HS-5011 (generated); owner's drawings ISO-HS-5011 | | |
| **Permits** | General work permit | | |

## Reason

Hose-station take-off valve at the water header seized.

## Scope of work

1. Isolate per the isolation list, drain and vent 1"-P150-HS-5011 between the isolations.
2. Unbolt and remove VALVE, BALL, FLOATING 1" P150 (1"-P150-HS-5011/A3) at E 62'-6"  N -4'-0"  EL 123'-7-1/2".
3. Inspect both flange faces (SW-NDE-5); dress or report damage.
4. Install a new VALVE, BALL, FLOATING 1" P150 of the same size, class and face-to-face (BALL1-P150), with new gaskets and studs, stem orientation as removed.
5. Tighten the studs to the stated preload in three passes, cross pattern (SW-WO-5).
6. Remove blinds, restore the line-up, leak test the two joints on start-up (SW-TEST-2).

Affected items: 1"-P150-HS-5011/A3, 1"-P150-HS-5011/A2, 1"-P150-HS-5011/A5

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-7

## Owner's isometric drawings (SW-IMP-6)

| Drawing | Line drawn | Work line | Sheets | Match to the plant model |
|---|---|---|---|---|
| ISO-HS-5011 | 1"-P150-HS-5011 | 1"-P150-HS-5011 | 1 | agrees with the model |

ISO-HS-5011 (PDF): ../isometrics-imported/ISO-HS-5011.pdf — the generated isometrics follow the ITP.

## Assets in the plant model (SW-WO-7)

| Tag | Kind | Description | Line | Location |
|---|---|---|---|---|
| rack-bent:720 | structure | Pipe rack bent at E 60'-0" | | E 60'-0"  N 0.000"  EL 122'-0" |
| 1"-P150-HS-5011/A3 | item | VALVE, BALL, FLOATING 1" P150 | 1"-P150-HS-5011 | E 62'-6"  N -4'-0"  EL 123'-7-1/2" |
| 1"-P150-HS-5011/A2 | item | GASKET 1" P150 COMPRESSED SHEET | 1"-P150-HS-5011 | E 62'-6"  N -4'-0"  EL 123'-3/4" |
| 1"-P150-HS-5011/A5 | item | GASKET 1" P150 COMPRESSED SHEET | 1"-P150-HS-5011 | E 62'-6"  N -4'-0"  EL 124'-2-1/4" |
| 3"-P150-W-2008/T:1"-P150-HS-5011 | item | TEE REDUCING 3" × 1" P150 | 3"-P150-W-2008 | E 62'-6"  N -4'-0"  EL 122'-6" |
| 1"-P150-HS-5011/B3 | item | VALVE, BALL, FLOATING 1" P150 | 1"-P150-HS-5011 | E 62'-6"  N 11'-0"  EL 104'-2-1/8" |
| 1"-P150-HS-5011/B6 | item | FLANGE WELD-NECK 1" P150 RAISED FACE | 1"-P150-HS-5011 | E 62'-6"  N 11'-0"  EL 104'-11-1/16" |

## Isolation list (SW-WO-3)

| # | Kind | Tag | Line | Action | Location | Fitted ✓ | Removed ✓ |
|---|---|---|---|---|---|---|---|
| 1 | equipment | 3"-P150-W-2008/T:1"-P150-HS-5011 | 3"-P150-W-2008 | No valve on this side within the line: isolate or depressure the parent line 3"-P150-W-2008 at its tee under the shutdown isolation | | | |
| 2 | valve | 1"-P150-HS-5011/B3 | 1"-P150-HS-5011 | Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-HS-5011/B3) | E 62'-6"  N 11'-0"  EL 104'-2-1/8" | | |
| 3 | blind | 1"-P150-HS-5011/B6 | 1"-P150-HS-5011 | Fit a line blind between 1"-P150-HS-5011/B3 and FLANGE WELD-NECK 1" P150 RAISED FACE (1"-P150-HS-5011/B6), work side | E 62'-6"  N 11'-0"  EL 104'-8-15/16" | | |

## Parts

| Description | Qty | Replaces |
|---|---|---|
| VALVE, BALL, FLOATING 1" P150 | 1 | 1"-P150-HS-5011/A3 |
| GASKET 1" P150 COMPRESSED SHEET | 1 | 1"-P150-HS-5011/A2 |
| STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150 | 1 | 1"-P150-HS-5011/A4 |
| GASKET 1" P150 COMPRESSED SHEET | 1 | 1"-P150-HS-5011/A5 |

| Description | Qty | Replaces |
|---|---|---|
| STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150 | 1 | 1"-P150-HS-5011/A7 |

## Inspections

| Id | Inspection | Criteria | Basis |
|---|---|---|---|
| WO-500-04-I1 | Flange faces of both joints before the new gaskets go in | No radial scratch across the sealing face (SW-NDE-5) | SW-NDE-5 |

## Bolt-up (SW-WO-5)

| Joint | Studs | Preload per stud (lbf) | Passes | Done ✓ |
|---|---|---|---|---|
| 1"-P150-HS-5011/A2 | 4 × 0.5" studs (1"-P150-HS-5011/A4) | 8,836 | 3 | |
| 1"-P150-HS-5011/A5 | 4 × 0.5" studs (1"-P150-HS-5011/A7) | 8,836 | 3 | |

## Test — Service leak test

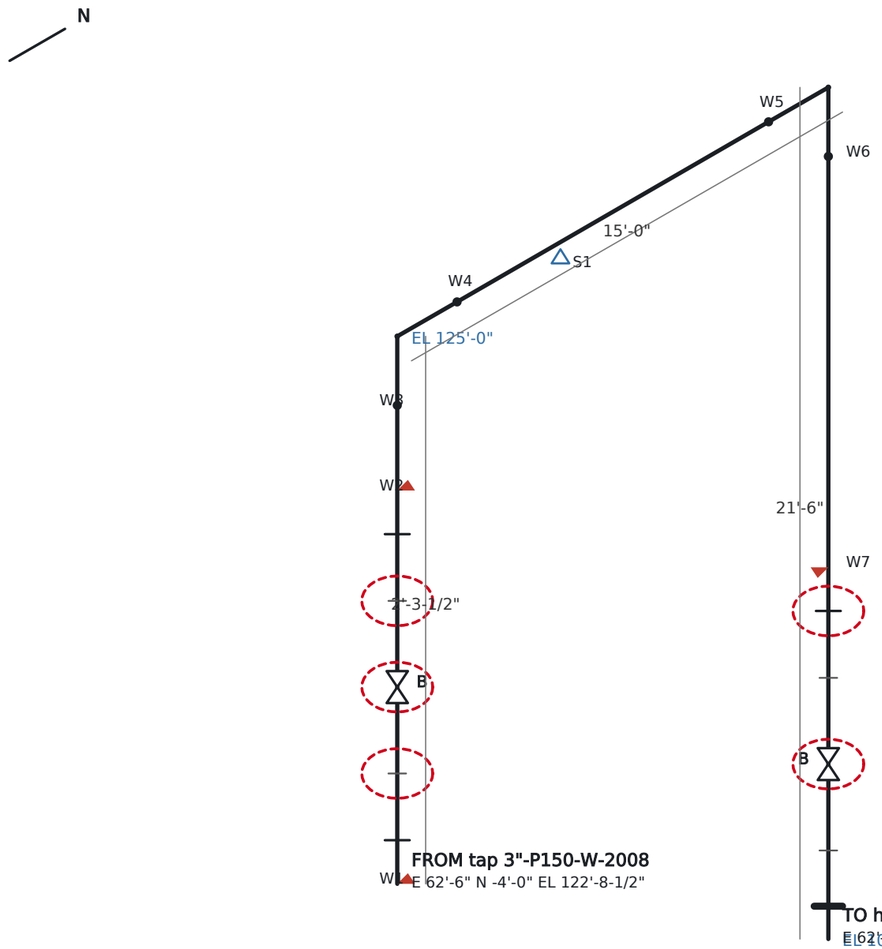
| | |
|---|---|
| **Description** | Both remade joints checked at operating pressure on start-up |
| **Basis** | SW-TEST-2 |

## Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

| # | Activity | Reference | Acceptance | Record | Contr. | Owner | 3rd party | Status | Signed (contr. / owner / 3rd) | Date | Remark |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained | SW-WO-3, isolation list | Every isolation in place and tagged; zero pressure at the vent | Signed isolation certificate | H | H | — | | | | |
| 2 | Valve 1"-P150-HS-5011/A3 removed; flange faces inspected | SW-NDE-5 | No radial scratch across the sealing face | Flange face inspection record | W | W | — | | | | |
| 3 | New valve verified: type, size, class, face-to-face and tag | BALL1-P150 | Matches the parts list and the line class | Material receiving record | R | R | — | | | | |
| 4 | New gaskets fitted, studs tightened to preload in three passes | SW-WO-5 | Preload per stud as stated; three passes, cross pattern | Bolt-up record per joint | W | W | — | | | | |
| 5 | Service leak test of the remade joints on start-up | SW-TEST-2 | No leak at either joint | Leak test record | W | W | — | | | | |
| 6 | Blinds removed, isolations returned to normal line-up per the isolation list | SW-WO-3 | Every blind removed and accounted for | Isolation certificate closed | H | H | — | | | | |
| 7 | Records complete; work order handed back to operations | SW-ITP-3 | All ITP activities accepted or not applicable | Signed ITP and completion record | H | H | R | | | | |

ISOMETRIC — 1"-P150-HS-5011 — Replace VALVE, BALL, FLOATING 1" P150 (1"-P150-HS-5011/A3)




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | FLANGE WELD-NECK 1" P150 RAISED FACE | 3 |
| 2 | GASKET 1" P150 COMPRESSED SHEET | 4 |
| 3 | VALVE, BALL, FLOATING 1" P150 | 2 |
| 4 | STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150 | 4 |
| 5 | PIPE 1" P150 BEVELLED ENDS | 34'-9-11/16" |
| 6 | ELBOW 90° R = 5/3 OD 1" P150 | 2 |
| 7 | FLANGE BLIND 1" P150 RAISED FACE | 1 |

**WELDS 7 (3 FIELD) · SUPPORTS 1 · SPOOLS 4**

W1F W2F W3 W4 W5 W6 W7F

## 1"-P150-HS-5011

CLASS W150 — Water (cooling and process)
SERVICE HS · 1" · P150 · DESIGN 150 psi
tap 3"-P150-W-2008 → hose station
PIPE 34'-9-11/16" · WEIGHT 120 lb

**DWG ISO-1"-P150-HS-5011 REV 0 WO-500-04**

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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